ATOMIC STRUCTURE

1. Types of EM radiations

$\longrightarrow \lambda$

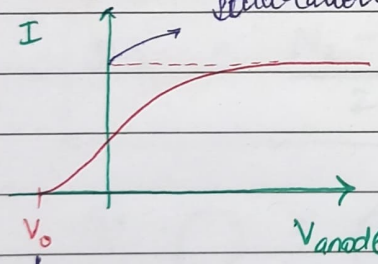
Cosmic < Gamma < X-rays < U.V < Visible < Infrared < Micro < Radio

(Call the Gentleman in his X U V, 9 am Ready)

2. $h\nu = \omega_0 + KE_{max}$

→ on ↑ freq. ⇒ KE of photoelectron ↑.
→ on ↑ intensity ⇒ no. of " " ".

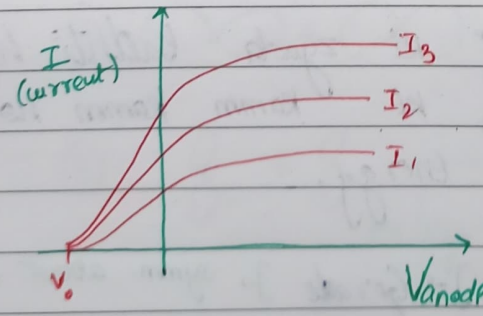
3. $E(\text{in eV}) = \dfrac{12400}{\lambda(\text{in } A^\circ)}$

4. 

$eV_0 = KE_{max} = h\nu - h\nu_0$

$V_0 = \dfrac{hc}{\lambda e} - \dfrac{\omega_0}{e}$

↳ stopping potential → mim^n Potential to stops the photo e$^-$ effect

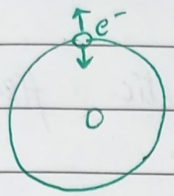
5. 

Intensity $I_3 > I_2 > I_1$

(* V_0 is dependent only on freq.)

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6. Bohr's Model



$$F_{orbital} = F_{Bohr}$$

$$\frac{kze^2}{r^2} = \frac{mv^2}{r} \quad -①$$

Bohr's postulate (Angular $\vec{p}$ is quantized) $\Rightarrow$ $mvr = \frac{nh}{2\pi}$ $-②$

Solve ① and ② for $\vec{v}$, r, and T.E.

$$v = \frac{2\pi kze^2}{nh} = v_0\left(\frac{z}{n}\right) \rightarrow 2.18 \times 10^6 \text{ m/s}$$

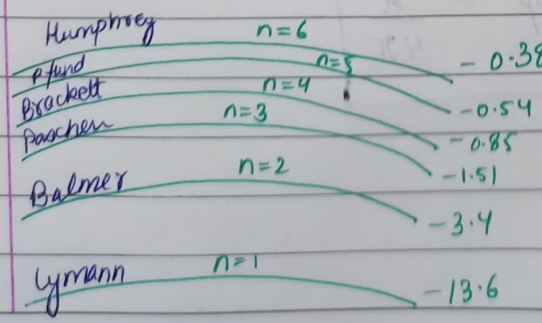
$$r = \frac{n^2h^2}{4\pi^2 kze^2 m} = r_0\left(\frac{n^2}{z}\right) \rightarrow 0.529\ A^\circ$$

$$\text{Time period} = \frac{n^3h^3}{4\pi^2 k^2 z^2 e^4 m} = T_0\left(\frac{n^3}{z^2}\right) \rightarrow 1.51 \times 10^{-16} \text{ sec}$$

$$T.E = -\frac{2\pi^2 k^2 z^2 e^4 m}{n^2h^2} = E_0\left(\frac{z^2}{n^2}\right) \rightarrow -13.6 \text{ eV} = -2.18 \times 10^{-18} \text{ J}$$

$$TE = \frac{PE}{2} = -KE$$

7.



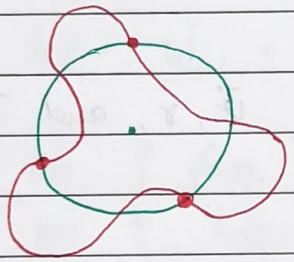
for H^+ $\Rightarrow$ lymann $\rightarrow$ U.V
Balmer $\rightarrow$ visible
↓Pas. $\rightarrow$ I.R

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8. Stark Effect → splitting of spectral lines under electric field.
≈ spark = electric

Zeeman Effect → " " " " " magnetic field.
↳ $\vec{B}$

9. De Broglie → $\lambda = \frac{h}{mc} = \frac{h}{mv} = \frac{h}{p}$
matter wave



$2\pi r = n\lambda$
↳ radius of orbit ↳ e^- wave

$\Rightarrow mvr = \frac{nh}{2\pi}$

$\lambda = \frac{h}{\sqrt{2KE \cdot (m)}}$

10. If an e^- is acc. btw a Potential diff of x Volt.
(i.e, given an energy of x eV) ↳ $\lambda = \sqrt{\frac{150}{x}}$ (A°)

If α particle → $\lambda = \frac{0.101}{\sqrt{x}}$

11. Heisenburg Uncertainity Principle :

$$\Delta x \cdot \Delta p \geq \frac{h}{4\pi}$$

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12. Quantum Nos.

a) (n) Principal (1, 2, 3 - - -) → size + dist from nucleus — Shell (1, 2, 3 - - -)

b) (ℓ) Azimuthal (0, 1, 2 - - -) → shape + energy — Subshell (s, p, d - - -)
(AKA Angular mom. & no.)

c) (m) Magnetic (−ℓ to +ℓ) → orientation — orbital (p_x p_y p_z - - -)

d) (s) Spin (+½ or −½) — spin (↑ or ↓)

13. a) no. of e⁻s in each shell ⇒ $2n^2$

b) no. of e⁻s in each subshell ⇒ $2(2\ell+1)$

c) Angular momentum in an orbital ⇒ $\sqrt{\ell(\ell+1)}\,\frac{h}{2\pi}$

d) Magnetic Moment (spin) → $\mu = \sqrt{n(n+2)}$ Bohr magneton
↳ no. of unpaired e⁻s.

14. i) Aufbau's Principle
↳ e⁻s are filled s.t net energy ↓.
(n+ℓ) rule → (n ℓ) ↓ ⇒ energy ↓
(n+ℓ) same ⇒ jiska 'n' ↑ → E ↑

ii) Pauli
↳ no 2 e⁻s can have all 4 quantum nos. same
↳ 1 orbital → max 2 e⁻s w/ opp. spin. ↑↑ ✗ ↑↓ ✓

(iii) Hund's
↳ In degenerate orbitals → each one first singly occupied.

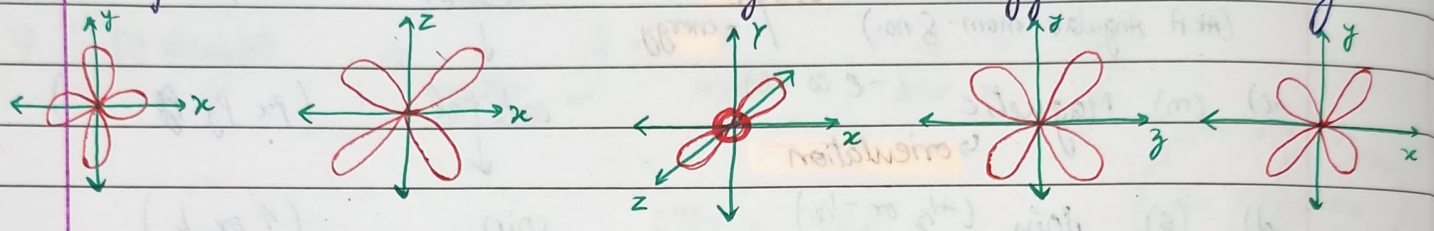
↑↓ | ↑ | | ✗

↑ | ↑ | ↑ ✓

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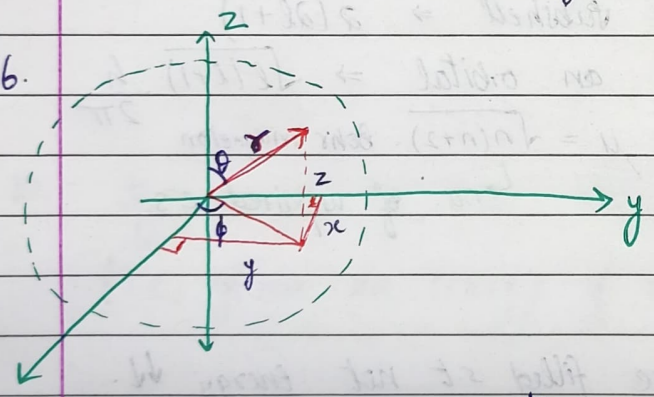
* 3d orbital ⇒ $n = 3$ $\ell = 2$ $m = -2$ to $+2$

−2	−1	0	1	2
$d_{x^2-y^2}$	d_{zx}	d_{z^2}	d_{yz}	d_{xy}



15. Nodes → spherical nodes = $n - \ell - 1$
Angular / planar = ℓ

16.



$\psi_{(r,\theta,\phi)} = R_{(r)}\ \Theta_{(\theta)}\ \Phi_{(\phi)}$

ψ → wavefunction

$R_{(r)}$ → radial part → depends on (n, ℓ)

$\Theta_{(\theta)}\ \Phi_{(\phi)}$ → Angular part → depends on (ℓ, m)

ψ → no significance

ψ^2 → probability of e^- density

Maal = (density of maal) × (vol. of maal)

Prob. of e^- → radial prob. distribution

density of maal → ψ^2

vol. of maal → $4\pi r^2 dr$